



Hyoree Kim

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Education

Mar. 2020 – Feb. 2026 **Sungshin Women’s University**, Seoul, Korea
School of AI Convergence, B.S.
GPA: 4.0 / 4.5

Research Interests

Computer Vision · Medical Imaging · Computational Pathology

Research Experiences

- Sep. 2025 – Feb. 2026 **Seoul National University Hospital**, BMI Lab
Research Intern
– Conducted computational pathology research on whole slide image (WSI) analysis, covering IHC tissue segmentation for immune cell quantification and H&E-based toxicologic pathology screening using foundation model transfer learning.
- Jun. 2023 – Dec. 2023 **Korea Institute of Science and Technology (KIST)**, Bionics Research Division, AIMI Lab
Research Intern
– Developed deep learning models for medical image analysis, including (1) hepatocellular carcinoma (HCC) segmentation in CT scans using the transformer-based UNETR model with aorta contrast referencing, and (2) breast cancer subtype classification in MRI incorporating protein marker labels (ER, PR, HER2, Ki-67) via a Vision Transformer architecture.
- Mar. 2023 – May. 2023 **Korea Institute of Science and Technology (KIST)**, Bionics Research Division, LIDS Lab
Research Intern
– Studied foundational deep learning concepts and academic papers focusing on medical image classification and segmentation, gaining insights about deep learning techniques and the application of deep-learning in healthcare.

Projects

- Sep. 2025 – Nov. 2025 **Seoul National University Hospital**
Endometrium ROI Segmentation in IHC WSI
– Developed an endometrium ROI segmentation model in IHC-stained WSIs as a prerequisite for automated immune cell (T cell, uNK cell) counting. Identified that the U-Net baseline (WSI Dice 0.78) relied on staining color rather than morphological features. Applied transfer learning with HEST-1k (NeurIPS 2024) fine-tuned DeepLabV3 weights to capture morphological features, combined with Stratified K-Fold and a 4-level Color Augmentation ablation study, achieving a final WSI Dice of 0.95.
- Nov. 2025 – Feb. 2026 **Seoul National University Hospital**
Few-Shot Hepatotoxicity Classification for Toxicologic Screening
– Led an industry-academia collaboration with medical AI company SPASS. In drug development, approximately 90% of rat liver tissue slides are normal, yet all require exhaustive manual review. Developed a negative screening model to automatically filter normal slides.
– Leveraged TANGLE (CVPR 2024) Pan-Cancer pretrained encoder (UNI + ABMIL), trained on 33 TCGA cancer types, as a frozen feature extractor to obtain slide embeddings from rat liver H&E WSIs, training only an MLP classifier on top for hepatotoxicity classification.
– Designed few-shot experiments anticipating limited labeled data for future multi-organ expansion. Achieved AUROC 0.909 and Recall 0.963 using only 15% of training data (935 slides), outperforming fully trained ABMIL from scratch (AUROC 0.859, Recall 0.836).

Feb. 2025 – Jun. 2025	<i>Zero-Shot Health Insight Generation from Tongue Images using MedGemma</i> – Built a small prototype using the public TongueDx dataset with MedGemma-4B-IT in a zero-shot setting to test whether a pretrained multimodal model can infer health-related cues from minimal inputs. Combined tongue images, age/sex, and brief captions to generate template-conformant reports with visual findings, reasoning, tongue-pattern labels, and nutrient recommendations.
Sep. 2023 – Dec. 2023	Korea Institute of Science and Technology (KIST) <i>Breast Cancer Classification Enhancement Using Protein Markers</i> – Improved breast cancer classification in MRI images through a Vision Transformer model incorporating multi-marker labels (ER, PR, HER2, Ki-67) alongside cancer presence. Demonstrated the feasibility of non-invasively identifying protein markers using MRI.
Jun. 2023 – Aug. 2023	Korea Institute of Science and Technology (KIST) <i>Improving HCC Diagnostic Precision in CT Using the UNETR Model</i> – Enhanced detection performance of HCC in CT scans by adjusting HCC contrast through aorta contrast referencing with the transformer-based UNETR model, enabling more precise HCC detection.
Mar. 2023 – May. 2023	Korea Institute of Science and Technology (KIST) <i>Image Classification on CIFAR-10 using ResNet</i> – Conducted image classification of the CIFAR-10 dataset, focusing on model optimization and accuracy improvement using the ResNet architecture.
Feb. 2023	<i>Breast Cancer Surgery Patient Survival Rate Prediction</i> – Employed various ML models (SVM, Random Forest, etc.) to predict survival rates of breast cancer surgery patients. SVM showed the best performance after normalization; Random Forest excelled when using the top five features.
Jan. 2023	<i>Lung Region Segmentation in CT Images</i> – Adjusted U-Net model parameters for lung segmentation in a CT dataset and verified segmentation performance by visualizing the results.

Awards and Honors

Sep. 2023	Internship Scholarship
Sep. 2022	Academic Excellence Scholarship
Mar. 2021	Academic Excellence Scholarship

Skills and Techniques

Programming	Python, Java, JavaScript
Frameworks	PyTorch/PyTorch Lightning, MONAI
Tools	GitHub, Docker